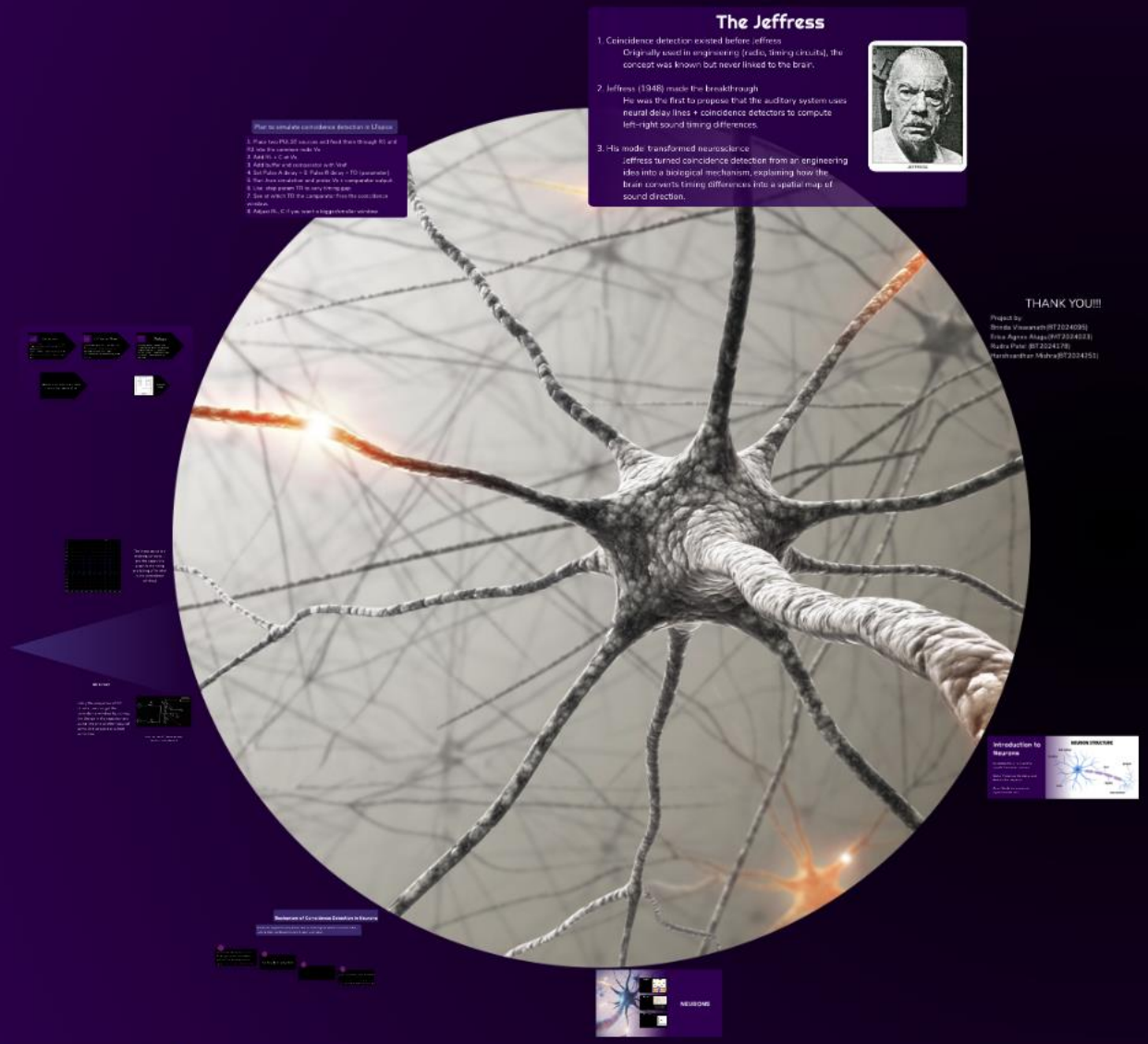


Temporal Coincidence Detector

A study of how neural systems detect near-simultaneous inputs using RC-based timing mechanisms.

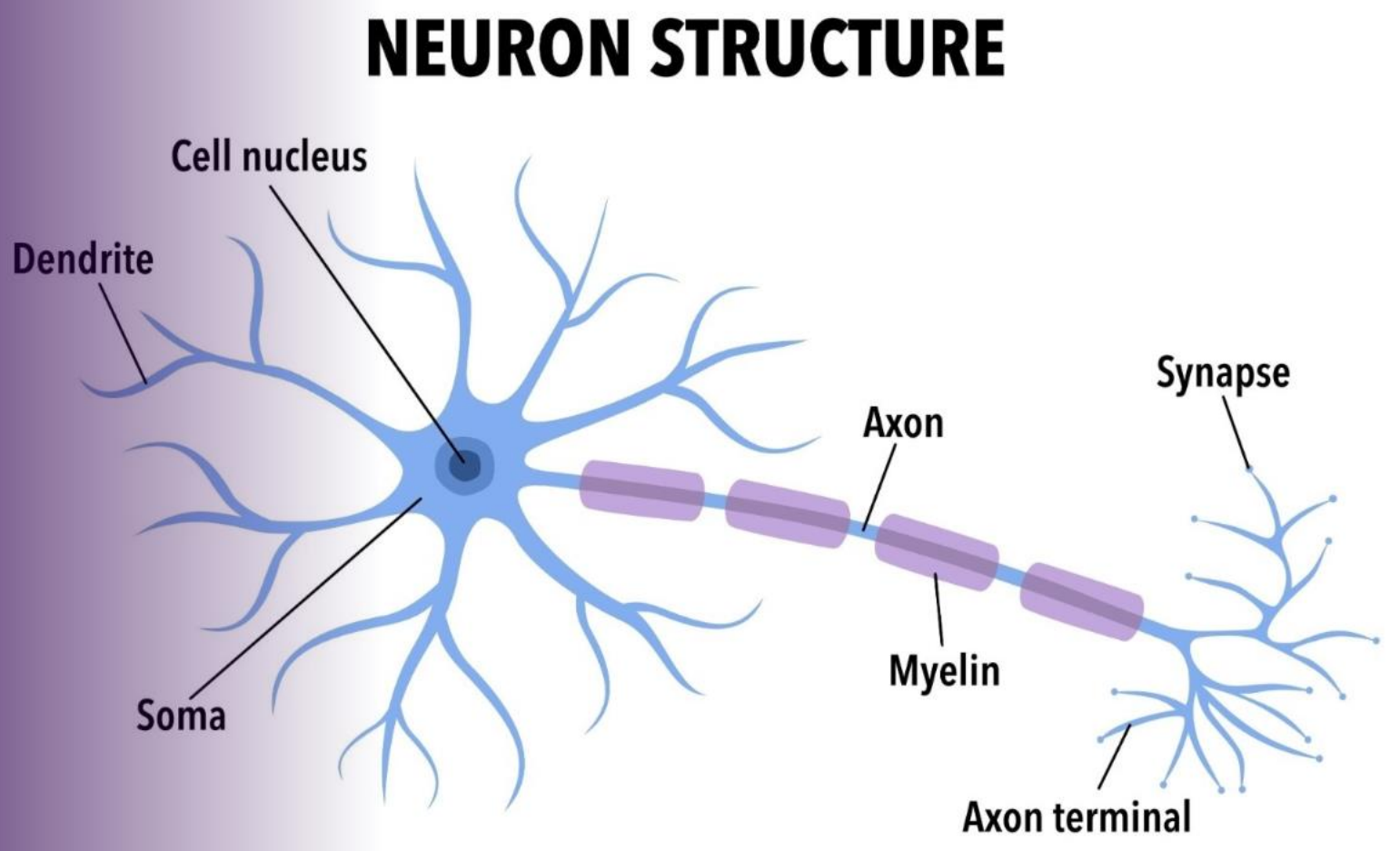
Prezi

Introduction to Neurons

Dendrites: Receive incoming signals from other neurons.

Soma: Processes the signals and decides the response.

Axon: Sends the processed signal to other cells.



Prezi

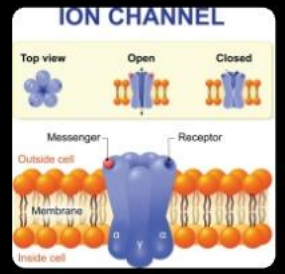
NEURONS

Channels

Membrane (Capacitor): The neuron's thin fatty layer that separates inside and outside charges, allowing it to store electrical potential.

Ion Channels (Resistors/Switches): Protein gates that open or close to let ions flow, controlling the neuron's electrical activity.

The Axon (The Wire): This transmits the electrical pulse



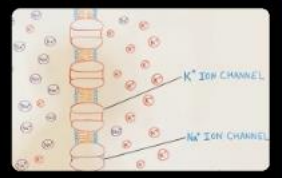
K + Na

Before a neuron receives any signal, it is "charged up" like a battery. This is called the Resting Membrane Potential (approx. -70 mV).

Two forces of Physics:
1)The Chemical Force (Diffusion/Entropy)
2)The Electrical Force (Electrostatic Pressure)

The Balance (Equilibrium):
The neuron reaches a standoff. Potassium tries to leave (Diffusion) but is pulled back (Electrostatic). The point where these two forces perfectly cancel each other out is the Equilibrium Potential. For a neuron, this settles at -70 mV

"Why is there a potential differance?"



$$E = \frac{RT}{zF} \ln\left(\frac{[Ion]_{out}}{[Ion]_{in}}\right)$$

All-or-Nothing

The Input (EPSP - Excitatory Postsynaptic Potential): Incoming signals allow a small amount of Sodium (Na+) to enter, slightly raising the voltage from its resting state (-70mV).

The Decision (-55mV): If the voltage reaches the Threshold of -55mV, "voltage-gated" channels physically snap open; if it fails to reach this number, the signal dies (Output = 0).

The Fire (Action Potential): Once the threshold is breached, a massive flood of Na+ rushes in, instantly spiking the voltage to +30mV (Output = 1).
"When does neuron fire a signal?"

Prezi

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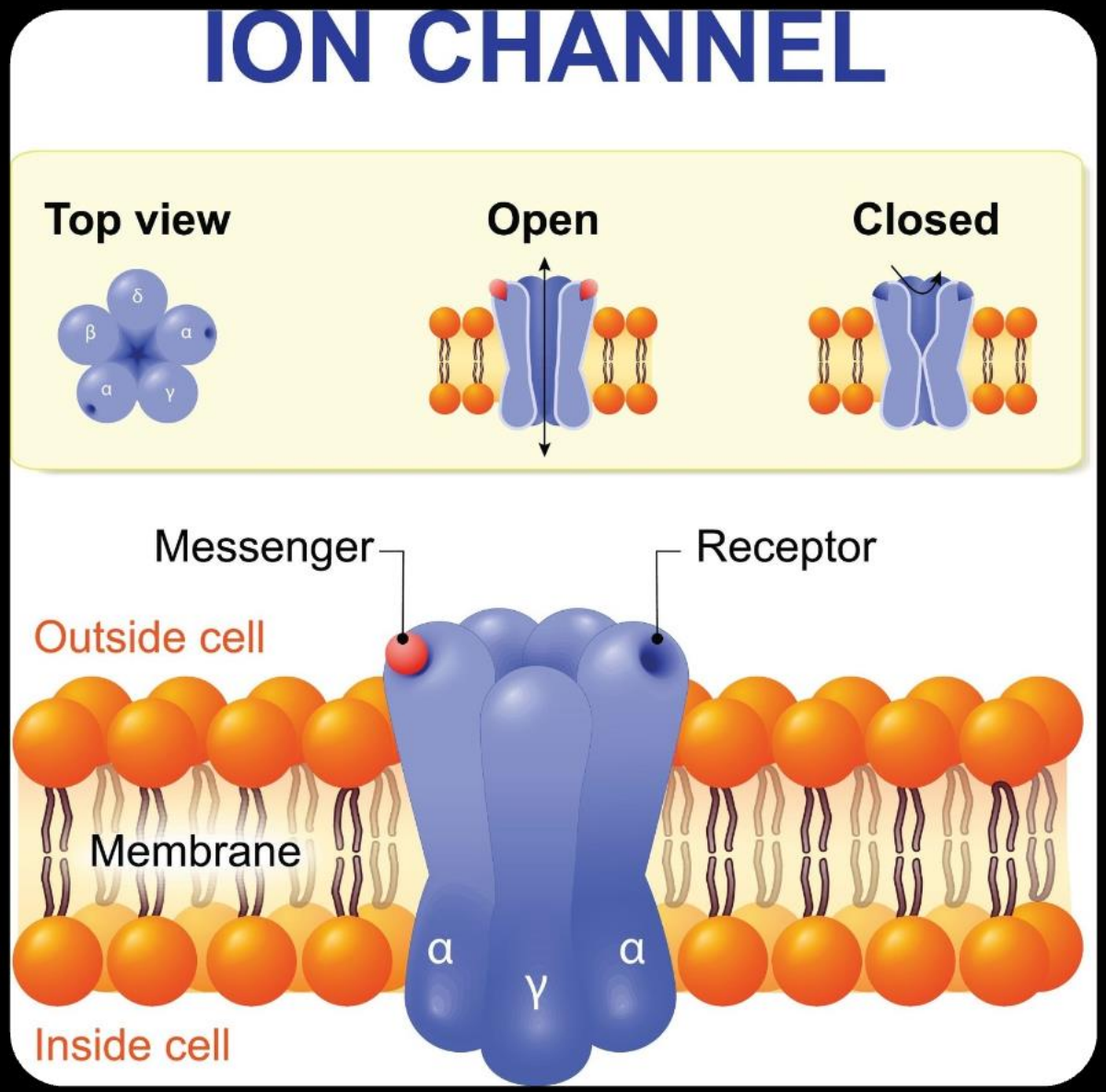

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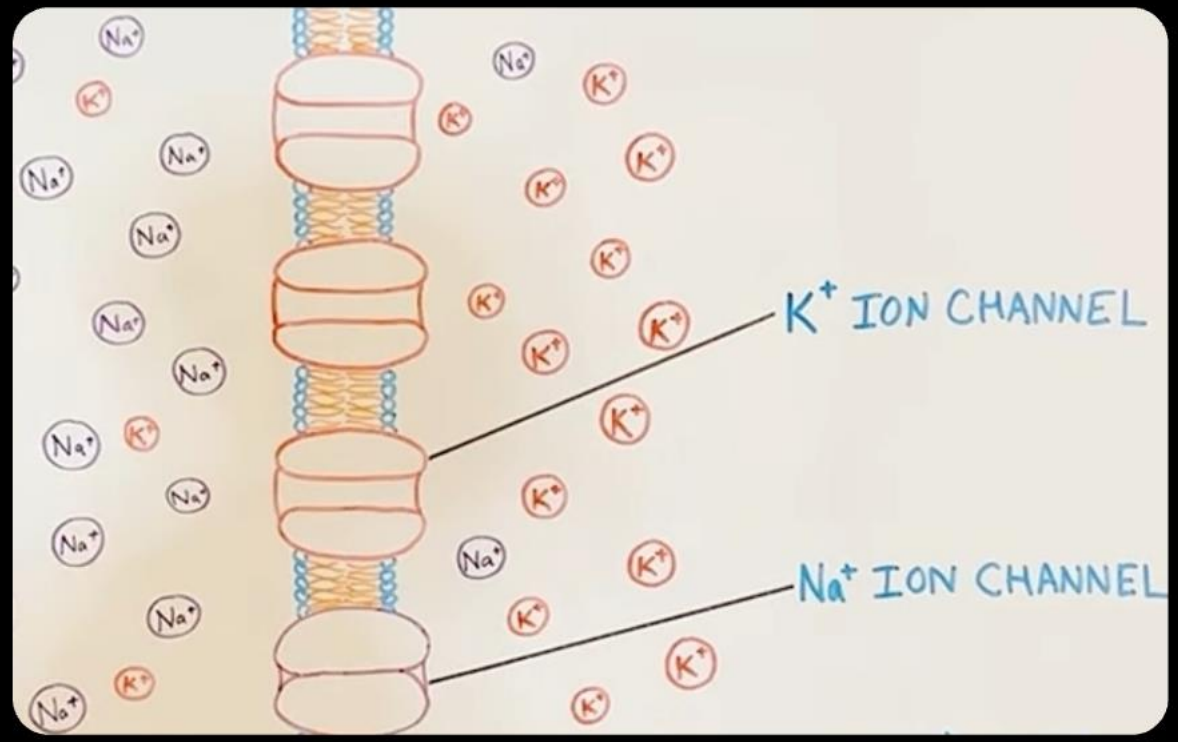

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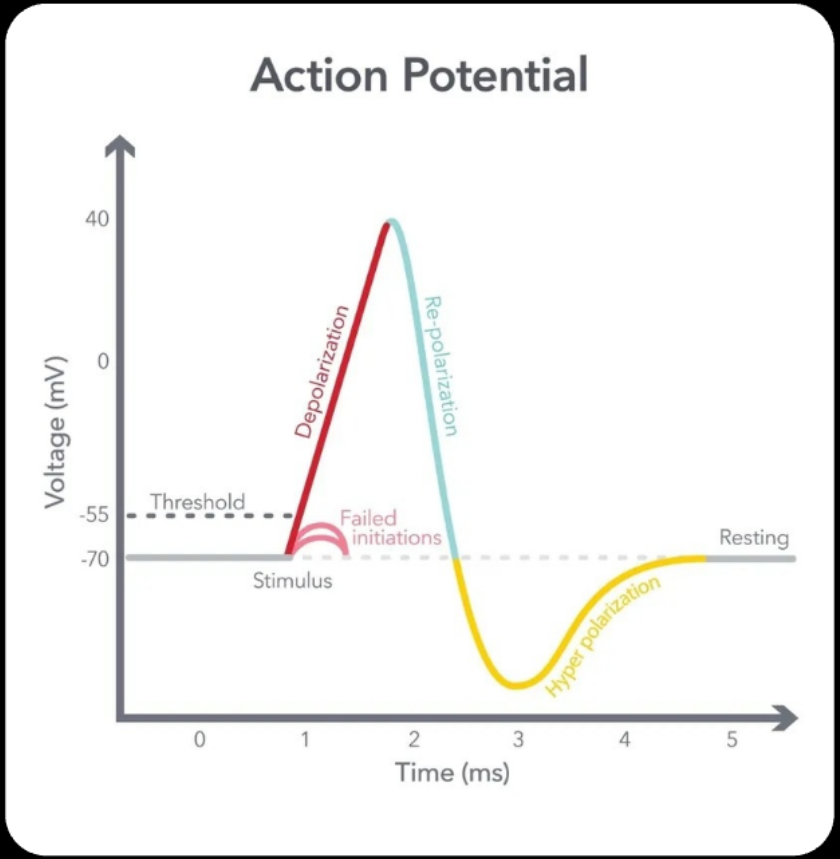

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Mechanism of Coincidence Detection in Neurons

A neuron responds only when two or more inputs arrive closely in time, letting their combined signals trigger activation.

Input: A signal hits the dendrite tip, creating a local electrical spike.
Decay: As the signal travels toward the center (Soma), electricity leaks out and fades.
Arrival: It reaches the Soma as a tiny ripple, too weak to act alone.
Result: Failing to reach the -55mV Threshold, the neuron ignores it. No Fire.

So, how do we solve this?

Case 1: The Failure using +10 mV as the input at the dendrite.

Travel & Decay: Input 1 enters the dendrite at +10 mV but shrinks to +8 mV by the time it reaches the Soma.

First Fail: The Soma voltage hits -62 mV (-70 + 8). This misses the -55 mV threshold. No Fire.

Reset: During the time gap, the voltage leaks back down to -70 mV.

Second Fail: Input 2 arrives later (also shrinking to +8 mV). Starting from zero, it also fails. No Fire.

Case 2: The Success (Temporal Coincidence).

First Hit: EPSP 1 travels to the Soma (shrinking to +8 mV). Voltage rises to -62 mV.
Coincidence: Before the first signal fades, EPSP 2 arrives immediately (also +8 mV).
The Sum: Since the first one didn't leak away yet, they stack up: -62 + 8 = -54mV.
Success: This crosses the -55 mV threshold. The trigger snaps open. FIRE!

Prezi

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Prezi

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Prezi

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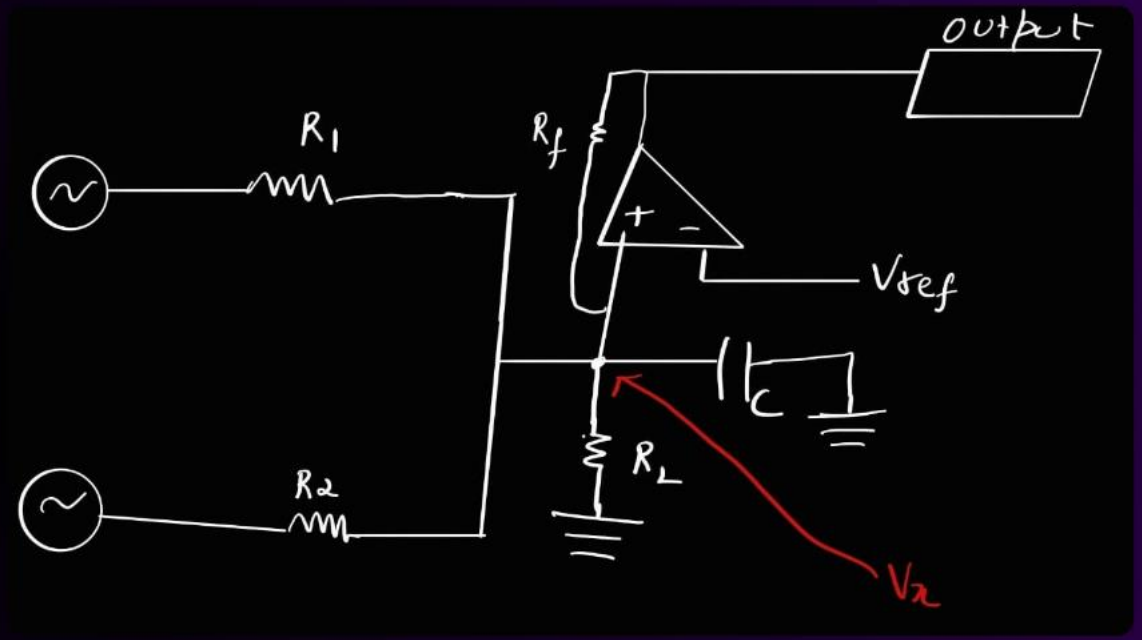
Prezi

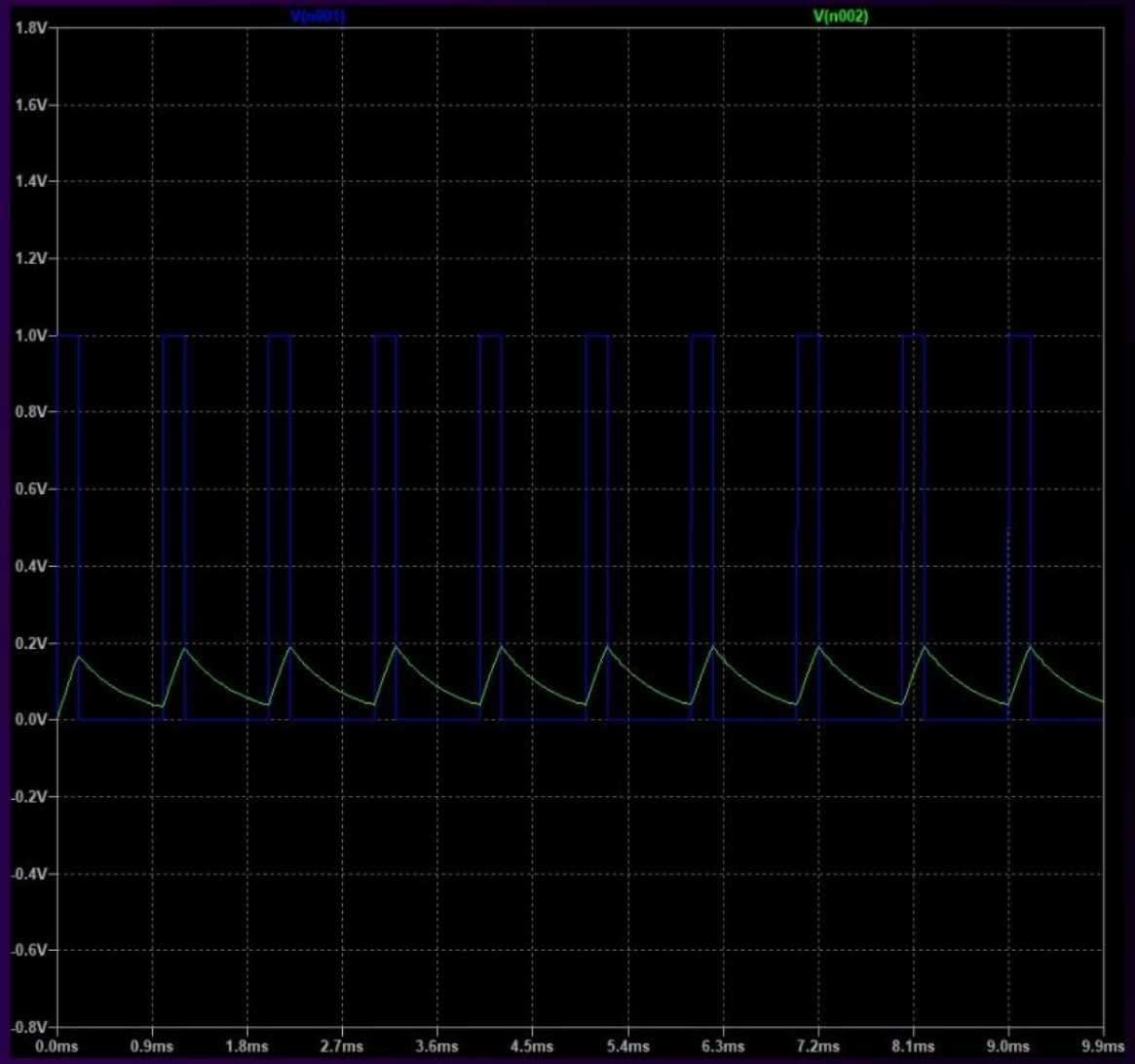
RC Circuit

Using the properties of RC circuits , we can get the coincidence window by storing the charge in the capacitor and using it to give another input of same kind at exact or almost same time



How can two RC paths be used to detect coincidence ?

Prezi



The input pulse is a rectangular wave , and the capacitive graph is the rising and falling of Vx (that is the coincidence window)

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Plan to simulate coincidence detection in LTspice

1. Place two PULSE sources and feed them through R1 and R2 into the common node Vx.
2. Add RL + C at Vx.
3. Add buffer and comparator with Vref.
4. Set Pulse A delay = 0, Pulse B delay = TD (parameter).
5. Run .tran simulation and probe Vx + comparator output.
6. Use .step param TD to vary timing gap.
7. See at which TD the comparator fires the coincidence window.
8. Adjust RL, C if you want a bigger/smaller window.

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The Jeffress

1. Coincidence detection existed before Jeffress

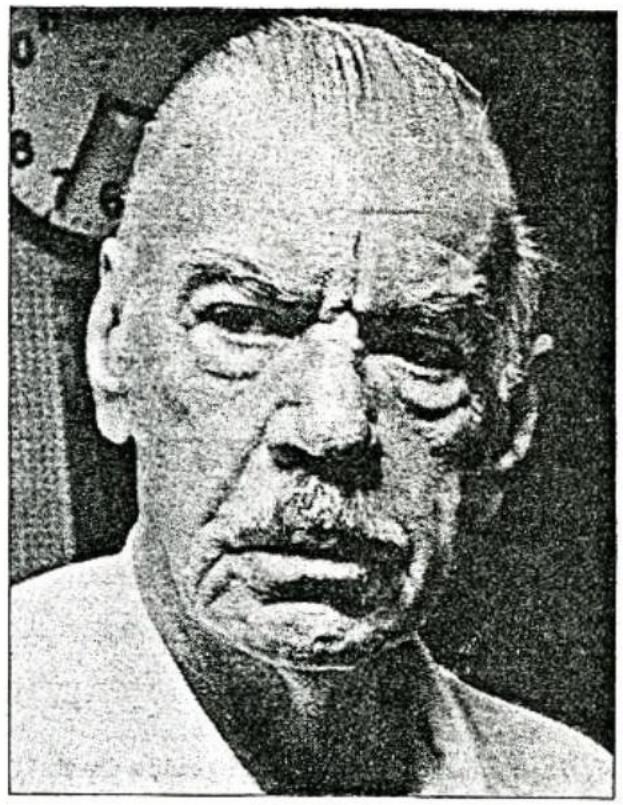
   Originally used in engineering (radio, timing circuits), the concept was known but never linked to the brain.

2. Jeffress (1948) made the breakthrough

   He was the first to propose that the auditory system uses neural delay lines + coincidence detectors to compute left–right sound timing differences.

3. His model transformed neuroscience

   Jeffress turned coincidence detection from an engineering idea into a biological mechanism, explaining how the brain converts timing differences into a spatial map of sound direction.



JEFFRESS

Prezi

Click Experiment

Trimble in 1928 showed the brain hears timing by sending two identical clicks were sent to each ear with a tiny delay.
Listeners heard one single click shifted left or right, depending on which ear got the click first — proving the brain detects interaural time difference (ITD) directly.

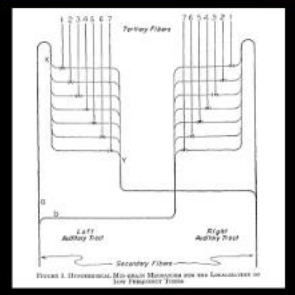
It's Time not Phase

This ruled out phase as the main cue, Because even when no phase cue existed (just two brief clicks), localization still shifted perfectly with timing.

Delays

Neural signals from the ears travel through fibers of different lengths and conduction speeds, so the auditory pathway naturally introduces tiny time delays before the signals reach the brainstem.

How do our ears detect that sound is coming from right or left ear?



The Jeffress Model

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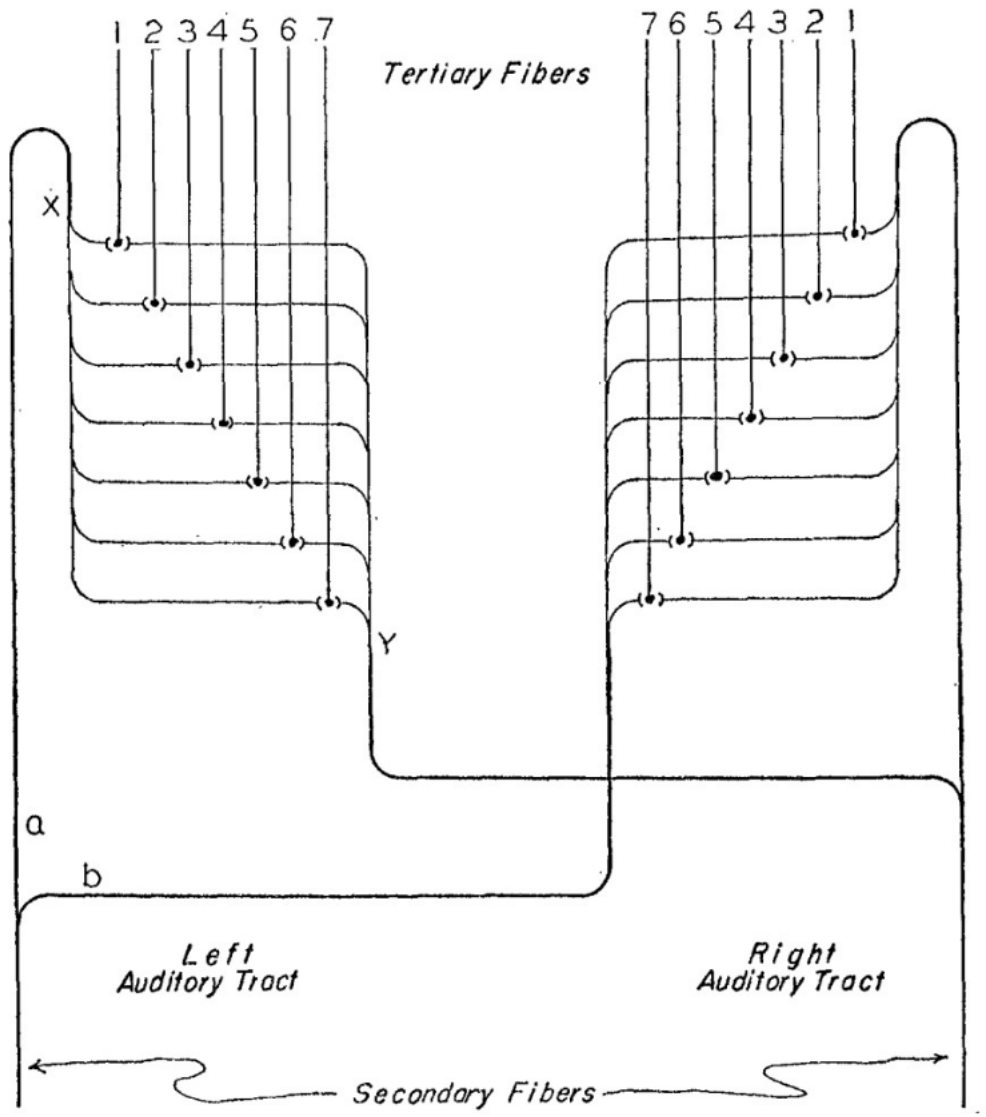

FIGURE 1. HYPOTHETICAL MID-BRAIN MECHANISM FOR THE LOCALIZATION OF LOW FREQUENCY TONES

The Jeffress Model

Prezi

THANK YOU!!!

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Harshvardhan Mishra(BT2024251)

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